PES1UG23AM225

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RAGHAV BUDUR

Warehouse Inventory & Pallet Location Tracker

1. Problem Overview

The Warehouse Inventory & Pallet Location Tracker is an automated warehouse management system designed to track pallet locations within warehouse racks, validate storage capacity limits, record stock movements, and support inventory management through barcode/RFID-based identification. The system improves warehouse efficiency, inventory accuracy, and storage safety.

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2. Actors

Primary Actor

Warehouse Operator

Responsible for scanning pallets, assigning storage locations, retrieving pallets, and recording stock movements.

Secondary Actor

Logistics Supervisor

Responsible for searching pallet locations, monitoring warehouse inventory, and generating inventory reports.

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3. Use Cases

1. Scan Pallet
2. Assign Pallet Location
3. Validate Weight Capacity
4. Retrieve Pallet
5. Log Stock Movement
6. Search Pallet Location
7. Generate Inventory Report
8. Send Low Stock Alert

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4. Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Core	The system shall scan pallets using barcode/RFID technology before any inventory operation is performed.	High	Pass: Pallet details are retrieved successfully after scanning. Fail: Scan data is not captured.	Ensures accurate identification of inventory items.
FR-002	Core	The system shall assign pallets to warehouse storage locations.	High	Pass: Valid storage location is assigned successfully. Fail: Storage assignment fails.	Enables organized warehouse storage.
FR-003	Core	The system shall validate rack weight capacity before confirming pallet placement.	High	Pass: Placement allowed only when weight limits are respected. Fail: Overloaded rack assignment is accepted.	Prevents unsafe storage conditions.
FR-004	Data	The system shall record stock movements between warehouse locations.	Medium	Pass: Movement details are stored in the system log. Fail: Movement occurs without a log entry.	Maintains inventory traceability.
FR-005	Reporting	The system shall generate inventory reports and support pallet location searches.	Medium	Pass: Correct inventory data and pallet locations are displayed. Fail: Incorrect information is returned.	Supports inventory monitoring and decision-making.

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5. Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The system shall return pallet location search results within 100 milliseconds.	High	Pass: Average search response time is less than or equal to 100 ms.	Ensures efficient warehouse operations.
NFR-002	Reliability	The system shall maintain accurate inventory records with at least 99.9% data consistency.	High	Pass: Inventory audits show no significant discrepancies.	Ensures reliability of warehouse data.

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6. UML Use-Case Diagram Description

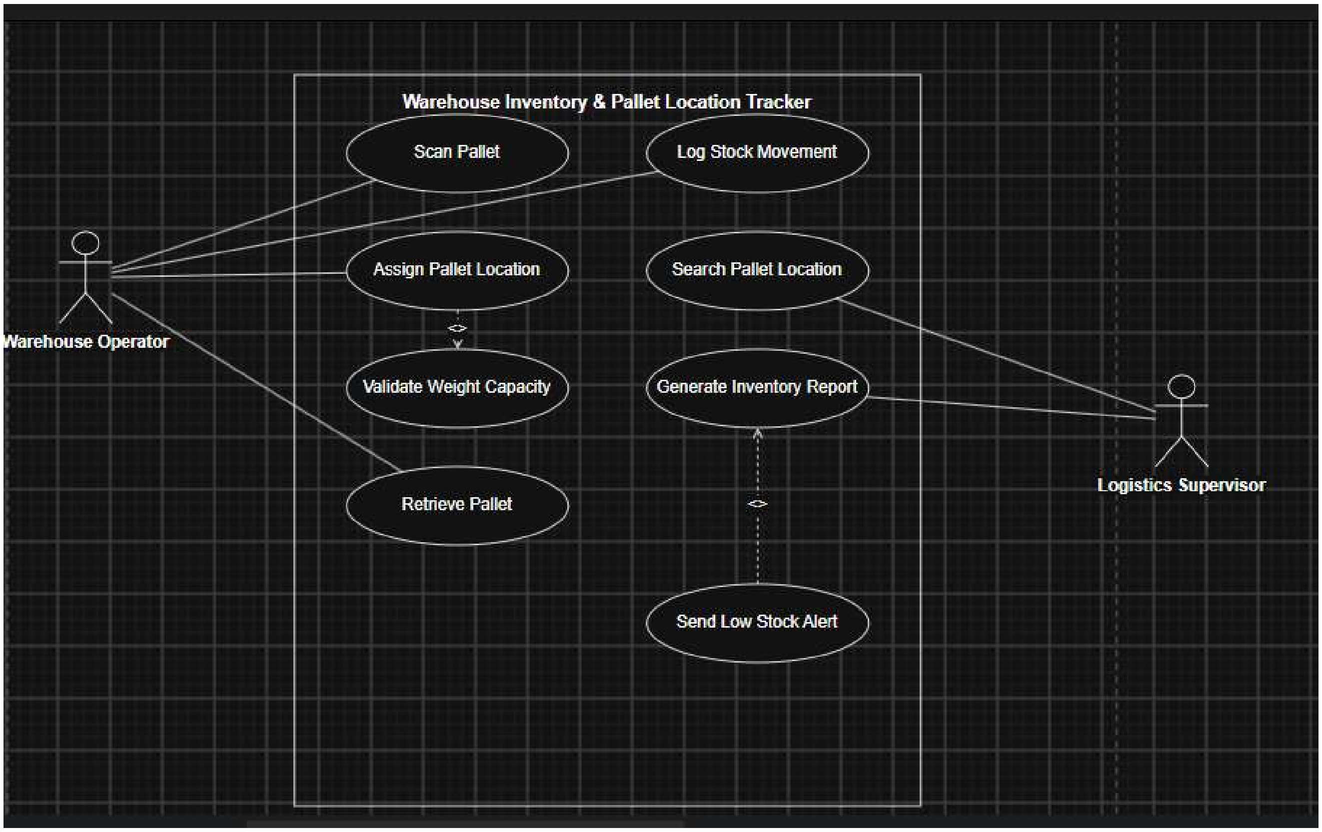
Actors

Warehouse Operator

- Scan Pallet
- Assign Pallet Location
- Retrieve Pallet
- Log Stock Movement

Logistics Supervisor

- Search Pallet Location
- Generate Inventory Report



Relationships

Assign Pallet Location → Validate Weight Capacity

Reason: Every pallet assignment must validate rack capacity before storage is confirmed.

Send Low Stock Alert → Generate Inventory Report

Reason: Low-stock alerts are generated only when inventory levels fall below a predefined threshold.

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7. Use-Case Flow Specification

Use Case ID

UC-002

Use Case Name

Assign Pallet Location

Primary Actor

Warehouse Operator

Goal

Assign a pallet to a valid storage location while ensuring rack weight limits are not exceeded.

Preconditions

1. Warehouse Operator is authenticated.
2. Pallet has been scanned successfully.
3. Rack information exists in the warehouse database.

Postconditions

1. Pallet is assigned to a storage location.
2. Inventory records are updated.
3. Storage transaction is logged successfully.

Main Success Scenario

1. Warehouse Operator scans the pallet.
2. System retrieves pallet information.
3. Operator selects a storage location.
4. System validates rack weight capacity.
5. System confirms that capacity limits are not exceeded.
6. Pallet is assigned to the selected rack location.
7. Inventory records are updated.
8. Confirmation message is displayed to the operator.

Alternate Flow – Capacity Exceeded

1. Operator selects a storage location.
2. System detects that the selected rack exceeds allowable weight limits.
3. System rejects the placement request.
4. Warning message is displayed.
5. Operator selects a different storage location.
6. System repeats the validation process.

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8. Business Rules

1. Every pallet must be scanned before storage or retrieval.
2. Rack weight limits shall never be exceeded.
3. All stock movements must be recorded in the system.
4. Inventory reports shall reflect real-time warehouse data.
5. Low-stock alerts shall be generated whenever inventory falls below configured threshold levels.